

RAKESH KUMAR

AI / GENAI ENGINEER

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SUMMARY

AI Engineer with hands-on experience building production-grade **RAG, LLM safety, and MLOps** systems, and applied research in **deep learning for security** resulting in two IEEE-format papers. Strong full-stack background (FastAPI, React, AWS) for shipping AI features end-to-end.

TECHNICAL SKILLS

NLP / GenAI: RAG & RAG Evaluation, LangChain, LangGraph, LangSmith, MCP, FAISS, Vector Search, Embeddings, Prompt Engineering, LLM Fine-tuning, LLM Gateway, LLM Evaluation, LLM Guardrails

ML / DL: PyTorch, NumPy, Pandas, Matplotlib, Scikit-learn, CNNs, R-CNN, Fast R-CNN, YOLO, RNNs, LSTMs, GRU, Transformers, Transfer Learning

AWS: EC2, S3, RDS, Lambda, IAM, VPC, CloudFront, Route 53, CloudWatch, ECS, ECR, Bedrock, SageMaker

Languages & Full Stack: Python, C++, JavaScript React.js, FastAPI, SQLAlchemy, Alembic

Dev Tools & Data: Git/GitHub, Docker, PostgreSQL, MongoDB, MySQL, Kafka, Redis

System Design: Distributed Systems, Caching, CDN, API Gateway, Event-Driven Architecture, SQL/NoSQL, Sharding & Replication, WebSockets, High Availability, Load Balancing, Message Queues, Rate Limiting

CS Fundamentals: Data Structures & Algorithms, DBMS, Operating Systems, OOP

EXPERIENCE

Full Stack Web Developer Intern

July 2023 – Dec 2023

Johnnette Technologies

- Developed full-stack web apps using React.js, Node.js, Express.js; built RESTful APIs for authentication/authorization with JWT session management and RBAC.
- Built reusable React.js component libraries, improving UI consistency and reducing front-end development time across product modules.

AI / ML PROJECTS

AI Knowledge Workspace – Enterprise SaaS RAG Platform

[GitHub](#)

Python, FastAPI, React, LangChain, LangGraph, PostgreSQL, Redis, FAISS, Docker, AWS

- Built an enterprise-grade **RAG platform** ingesting PDF, DOCX, and image sources, with hybrid retrieval, reranking, streaming responses, and **hierarchical conversation memory**.
- Designed **multi-level RAG evaluation and regression testing** across component, pipeline, and application levels to catch regressions from model, prompt, and pipeline changes.

CloudOps Sentinel – Enterprise Incident Response Self-RAG Copilot

[GitHub](#)

Python, FastAPI, LangChain, LangGraph, OpenAI, Pinecone, SQLite, LangSmith, Docker, DigitalOcean

- Built an agentic Self-RAG copilot for incident response: private runbook retrieval via Pinecone, OpenAI embeddings/LLMs, relevance grading, grounded generation, and web-search fallback for knowledge gaps.
- Designed a self-correcting LangGraph workflow with conditional retrieval, answer verification, bounded retries, SQLite incident memory, LangSmith observability, and Docker-based deployment and Observability with langsmith.

LLM Guardrails – Multi-Layer AI Safety Pipeline

[GitHub](#)

Python, FastAPI, DeBERTa-v3, Detoxify, Microsoft Presidio, spaCy, AsyncIO

- Built an async, cost-tiered guardrail pipeline (validation → PII → ML models → LLM) using DeBERTa-v3 for prompt-injection detection, Detoxify for toxicity, and Presidio + spaCy for PII, with short-circuit logic.
- Used `asyncio to_thread` for non-blocking transformer inference under load; architected for extensibility with post-generation checks and a planned LLM-as-judge hallucination layer.

RESEARCH

Hybrid Phishing URL Detection – DeepTCN, Meta-Stacking, Concept Drift Monitoring [GitHub](#)

Python, PyTorch, Scikit-learn, TF-IDF, Isolation Forest, SciPy

- Stacked SGD Classifier, DeepTCN + Attention (AUC: 0.9932), and Isolation Forest via a Random Forest meta-learner; achieved 96%+ accuracy & recall on 555K+ URLs, with zero-shot evaluation on the PhiUSIIL OOD dataset.
- Built an 80,008-dim sparse feature pipeline (TF-IDF char n-grams + lexical features) with CSR compression, reducing memory from 32GB to 200MB.
- Integrated real-time concept drift monitoring (Page-Hinkley, ADWIN, DDM) over 160K predictions with an active-learning loop for retraining; co-authored an IEEE-format paper with Dr. Narayanan Prashanth (SCOPE, VIT).

EDUCATION

Vellore Institute of Technology

M.Tech – CSE (AI & ML) | GPA: 8.38 / 10

Vellore, India

July 2025 – Present

KIET Group of Institutions

B.Tech – CS and Information Technology | CGPA: 7.3 / 10

Ghaziabad, India

Nov 2020 – July 2024